

KUNSAN AB, Korea, Republic ofWMO: **471410**Lat: **35.904N**Lon: **126.616E**Elev: **33**StdP: **14.68**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **43219****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.5	20.9	8.3	8.4	23.7	11.2	9.7	25.7	32.6	30.1	28.1	30.2	7.6	100	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.1	89.9	79.5	87.8	79.2	85.8	78.3	81.3	85.9	80.3	85.0	79.4	83.8	10.4	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.4	158.2	83.6	79.1	151.1	82.7	78.5	148.1	82.4	45.2	86.0	44.2	85.6	43.1	84.0	87.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	25.4	21.7	18.7	12.8	93.7	4.0	2.4	9.9	95.4	7.6	96.7	5.3	98.1	2.4	99.8	99.8
(5)				11.5	83.6	3.7	1.7	8.8	84.8	6.6	85.8	4.6	86.7	1.9	87.9	87.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.2	32.1	35.2	42.8	53.0	62.9	71.2	78.4	80.2	71.7	60.6	48.3	36.5
	DBStd	17.35	6.35	5.93	5.59	5.98	4.59	3.46	3.94	4.08	4.50	5.40	6.70	6.78
	HDD50	1773	556	415	233	35	0	0	0	0	0	3	111	420
	HDD65	4546	1020	834	688	363	95	2	0	0	4	154	502	884
	CDD50	4030	0	1	10	124	400	635	881	936	652	331	59	1
	CDD65	1328	0	0	0	2	30	187	416	471	205	17	0	0
	CDH74	9759	0	0	0	8	99	607	3370	4603	1034	37	1	0
	CDH80	2666	0	0	0	1	6	59	889	1547	163	1	0	0
Wind	WSAvg	8.1	8.6	9.1	9.7	8.9	7.9	7.1	7.9	7.6	6.9	7.4	8.3	8.5
Precipitation	PrecAvg	50.1	1.3	1.5	2.0	3.3	3.4	5.8	10.9	10.6	5.4	2.1	2.1	1.7
	PrecMax	70.7	3.1	4.0	4.3	9.6	5.9	13.3	22.5	22.6	23.0	5.4	6.9	3.0
	PrecMin	32.9	0.3	0.2	0.5	1.0	0.9	0.5	1.9	2.0	0.7	0.3	0.4	0.2
	PrecStd	10.2	0.7	1.0	1.0	2.0	1.4	3.4	5.0	5.6	4.8	1.6	1.5	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.8	58.7	64.4	73.8	80.3	84.6	92.6	94.7	87.4	77.3	69.4	57.1
		MCWB	46.7	53.6	55.4	61.5	65.6	72.0	79.5	79.7	75.6	65.6	61.1	50.7
	2%	DB	46.7	51.6	59.1	69.5	76.8	81.5	89.2	91.2	84.1	74.8	64.6	52.2
		MCWB	42.4	45.7	52.6	59.1	65.0	71.4	79.6	79.7	74.3	65.6	58.8	47.1
	5%	DB	44.2	46.9	55.5	66.0	73.8	79.1	86.4	89.3	81.8	71.9	62.0	49.7
		MCWB	40.3	42.5	49.3	57.9	64.1	71.0	79.0	79.5	72.9	63.7	57.0	45.6
	10%	DB	41.2	44.4	52.0	62.7	71.4	77.3	84.5	87.4	79.2	69.8	59.0	46.6
		MCWB	37.6	40.6	46.8	56.0	63.1	70.5	78.3	79.2	71.5	62.3	53.8	42.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.9	54.8	58.8	66.1	71.8	77.9	82.7	83.0	80.0	71.7	64.1	52.9
		MCDB	49.8	56.8	61.2	70.4	75.8	80.2	85.9	87.5	83.6	74.7	66.6	55.8
	2%	WB	43.2	46.7	54.4	62.3	68.3	75.4	81.0	81.8	77.5	68.4	60.6	48.6
		MCDB	45.7	49.8	57.3	66.3	72.4	77.7	85.2	86.6	80.9	71.8	62.9	51.3
	5%	WB	40.5	43.4	50.4	59.4	66.3	73.8	80.1	80.8	75.5	65.6	57.7	45.8
		MCDB	43.0	46.0	53.6	63.7	71.2	76.4	84.3	85.8	78.9	69.3	60.8	48.6
	10%	WB	38.1	40.9	47.7	57.0	64.5	72.3	79.2	79.9	73.6	63.7	55.0	43.2
		MCDB	40.7	43.5	51.3	61.0	69.2	75.3	83.1	84.8	77.7	67.9	58.4	46.1
Mean Daily Temperature Range	5% DB	MDBR	12.9	13.9	14.8	15.1	14.3	11.7	9.7	11.1	13.5	15.7	14.7	13.4
		MCDBR	15.5	18.1	19.8	19.4	18.3	14.1	12.0	13.6	15.2	17.2	17.2	16.5
		MCWBR	11.9	13.3	13.4	11.5	9.5	7.1	5.6	5.7	7.8	10.0	12.7	12.6
	5% WB	MCDBR	14.3	16.1	17.7	16.2	15.2	11.1	10.5	11.3	11.7	14.4	15.3	15.4
		MCWBR	12.1	13.0	13.8	11.6	9.7	7.3	6.0	5.8	7.5	10.2	12.9	13.1
Clear-Sky Solar Irradiance	taub	0.455	0.494	0.568	0.589	0.608	0.679	0.630	0.572	0.502	0.486	0.472	0.455	
	taud	1.923	1.861	1.721	1.721	1.720	1.736	1.663	1.819	1.963	2.078	1.982	1.959	1.935
	Ebn at Noon	219	228	223	228	226	210	220	220	230	240	228	213	209
	Edh at Noon	47	56	70	73	73	79	67	57	48	49	45	43	
All-Sky Solar Radiation	RadAvg	745	1013	1387	1611	1772	1619	1419	1461	1333	1164	792	651	
	RadStd	56	92	131	127	182	151	198	213	133	93	89	51	